



ENGG100 - Practical 3

Engineering Computing and Analysis (University of Wollongong)



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ENGG100 – Engineering Computing and Analysis

Practical 3 – Week 3

In this Practical class, you will work through parts of the first, second and third chapters of the subject textbook “Essential MATLAB for engineers and scientists” (7th Edition) by Brian H. Hahn and Daniel T. Valentine, 2019.

STEP 1 – Access the online textbook through the library’s portal.

- 1.1 Access the textbook using the same process as Practical 1. Access the PDF versions of Chapter 1, Chapter 2 and Chapter 3 as required. Note the objectives of each chapter are highlighted in orange.

STEP 2 – Section 1.2.1 – Using the Editor and running a script.

- 2.1 Read and follow the instructions in Section 1.2.1 – Using the Editor and running a script. Use this technique in your future work, to create a script that can be easily re-run.

STEP 3 – Examples from Sections 2.3.8 and 2.4

- 3.1 Read and follow the instructions in Section 2.3.8 – Structure Plan. When typing this program into MATLAB include additional lines labelling each part of the program such as **Input**, **Operations** and **Output** stages (as described at the end of the section).
- 3.2 Read the structure plan in Section 2.4 Vertical Motion under Gravity. Draw the flow chart for this structure plan. Follow the instructions for how to generate a program from your structure plan and save as a script. Make sure to include plenty of additional comment lines to keep track of what your program is doing.

STEP 4 – Chapter 3.1 – The Program Design Process.

- 4.1 Now open the Chapter 3 PDF file. Read through Section 3.1 The Program Design Process, taking care to note the 7 steps of the design process. Read and follow the instructions in Section 3.1.1 – The Projectile Problem, until you reach the end of Steps 6 & 7. Consider how you would perform the necessary evaluation (or testing) in Step 6 to ensure correctness of your program?
- 4.2 Read the structure plan on Pages 94 & 95 for the example of finding the solutions to a quadratic equation. Consider how you would normally find the solution to a quadratic equation? Draw the flow chart for this structure plan. Specifically, indicate where the structure plan requires decision boxes to determine what the required computation is. Be sure to keep this flow chart handy for next week in **Practical 4**.
- 4.3 Can you think of other solutions to mathematical problems that require conditional calculations (such as the use of **if-then-else** statements or **do-while-loops** or **for-loops**)? Discuss your answers to the above questions with your peers.